

Hayden Fuller
S26 Internetworking of things
Lab 1
02/13/26
P1)
<https://web1.ecse.rpi.edu/~fulleh/bio.txt>

P2)
RFID:

```
(lab1) hayden@fullehrpi:~/Desktop/Lab1 $ python3 RFIDfullehWRbyte.py
Ready to Write
Wrote
Ready to Read
id: 733001155397
byte: bytearray(b'Hello Hayden!0W00x000x000x000x000x000x000x000x00')
(lab1) hayden@fullehrpi:~/Desktop/Lab1 $ |
```

```
import RPi.GPIO as GPIO
from MFRC522_IOT import MFRC522_IOT
import time
```

```
reader = MFRC522_IOT()
```

```
try:
    text = bytearray(b'Hello Hayden!')
    text.extend([0x4F, 0x57, 0x4F])
    print("Ready to Write")
    reader.writebytes(text)
    print("Wrote")
    time.sleep(1.0)
    print("Ready to Read")
    id, textbytes = reader.readbytes()
    print("id: ", id)
    print("byte: ", textbytes)
    time.sleep(1.0)
```

```
finally:
    GPIO.cleanup()
```

DHT11:

```
(lab1) hayden@fullehrpi:~/Desktop/Lab1 $ python3 DHT11fulleh.py
time: 2026-02-13 14:13:18.991728
temp: 24
humidity: 22
time: 2026-02-13 14:13:21.139649
temp: 25
humidity: 15
time: 2026-02-13 14:13:23.287253
temp: 25
humidity: 14
```

```
import RPi.GPIO as GPIO
from DHT11_lib import DHT11
import time
import datetime
```

```
GPIO.setwarnings(False)
GPIO.setmode(GPIO.BCM)
GPIO.cleanup()
instance = DHT11(pin=17) #Pin 11
```

```
while True:
    result = instance.read()
    if result.is_valid():
        print("time: " + str(datetime.datetime.now()))
        print("temp: %d" % result.temperature)
        print("humidity: %d" % result.humidity)
    time.sleep(1)
```

HCSR04:

```
(lab1) hayden@fullehrpi:~/Desktop/Lab1 $ python3 HCSR04test.py
distance is: 50.78387260437012 cm
distance is: 50.5426287651062 cm
distance is: 50.64893960952759 cm
distance is: 51.17231607437134 cm
distance is: 52.76697874069214 cm
distance is: 56.128036975860596 cm
distance is: 293.8268184661865 cm
distance is: 293.8349962234497 cm
```

```
import RPi.GPIO as GPIO
```

```
import time
```

```
from HCSR04_lib import HCSR04
```

```
GPIO.setwarnings(False)
```

```
GPIO.setmode(GPIO.BCM)
```

```
GPIO.cleanup()
```

```
TRIG = 4
```

```
#ECHO = 17 # pin 11
```

```
ECHO = 27 # pin 13
```

```
GPIO.setup(TRIG, GPIO.OUT)
```

```
instance = HCSR04(TRIG_pin=TRIG, ECHO_pin=ECHO) # BCM17
```

```
instance.init_HCSR04()
```

```
while True:
```

```
    dist = instance.measure_distance()
```

```
    print("distance is: ", dist, " cm")
```